Numerical Supplement

Causal Trust Gating for Battery Dispatch in Distribution Networks under Degraded Measurements

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This supplement accompanies the executed rolling-origin dispatch assessment. All values are derived from the v3.0.0 archive. The complete per-hour records, source hashes and input profiles are available at https://github.com/Devchandrasen/tudt-f-smart-energy/releases/tag/v3.0.0. The v2.0.0 archive uses a different design and is retained only as historical material.

1 Configuration and chronological partitions

The same electrical limits, storage equations and tariff settings apply to the principal controllers. Profile-factor values and fitted parameters remain representation specific. Temporal blocks retain all conditions and seeds in paired aggregation.

Table 1: Chronological sample counts. Counts refer to valid current-target pairs.

Representation	Origin	Train	Calibration	Development
IEEE 33	Summer	2879	743	719
IEEE 33	Winter	5831	719	743
MV aggregate	Summer	2879	743	719
MV aggregate	Winter	5831	719	743
MV asset profiles	Summer	2879	743	719
MV asset profiles	Winter	5831	719	743

Table 2: Last next-hour target dates retained inside each partition.

Representation	Origin	Last train	Last calibration	Last development
IEEE 33	Summer	2016-04-30	2016-05-31	2016-06-30
IEEE 33	Winter	2016-08-31	2016-09-30	2016-10-31
MV aggregate	Summer	2016-04-30	2016-05-31	2016-06-30
MV aggregate	Winter	2016-08-31	2016-09-30	2016-10-31
MV asset profiles	Summer	2016-04-30	2016-05-31	2016-06-30
MV asset profiles	Winter	2016-08-31	2016-09-30	2016-10-31

The summer test starts are 1, 5, 9, 13, 17, 21, 25 and 28 July. Winter starts use the same dates in November and December. Each block has 72 commands and 24 hours of prehistory. Measurement seeds are 11, 29 and 47. Conditions are Clean, Missing, Delayed, Noisy, Bias, Disturbance, Reference outage, Reference noisy and Common bias. Fourteen configurations provide 46,656 evaluated hours per controller and representation. The complete total is 1,959,552 records.

All study checks use voltage 0.95–1.05 p.u. and thermal utilisation at most one, with tolerance 10^{-7}. The native SimBench bus bands include 0.90–1.10 and 0.965–1.055 p.u. Reported failures concern the imposed study band. Battery efficiencies are 0.94, state-of-charge limits are 0.15–0.90 and every block starts at 0.55. IEEE storage is 0.8 MW and 3.2 MWh at bus 17; SimBench storage is 2 MW and 8 MWh at bus 15. Buying prices are 70, 240 and 150 EUR/MWh during 00:00–06:59, 17:00–21:59 and other hours, respectively. Export receives 40 EUR/MWh and throughput costs 10 EUR/MWh.

2 Development cutoffs and coverage

Cutoffs use received-reference development scores only. Lower and upper targets are 85% authorisation and 50% full dispatch, respectively. Equality is retained, so ties can increase achieved coverage. Values below are fixed before test dispatch.

Table 3: Fitted score thresholds and conditional development coverage.

Representation	Origin	Score	τ_L	τ_H	Authorised (%)	Full (%)
Aggregate	Summer	Agreement	0.573030	0.898015	85.000	50.008
Aggregate	Summer	Composite	0.729444	0.929623	85.000	50.003
Aggregate	Summer	Without freshness	0.702175	0.900993	85.000	50.003
Aggregate	Summer	Without completeness	0.642444	0.915425	85.000	50.003
Aggregate	Summer	Without maturity	0.820741	0.968927	85.000	50.013
Aggregate	Summer	Without agreement	0.775079	0.937968	85.000	50.003
Aggregate	Winter	Agreement	0.570326	0.899086	84.997	50.003
Aggregate	Winter	Composite	0.698440	0.898524	84.997	50.003
Aggregate	Winter	Without freshness	0.643358	0.857825	84.997	50.003
Aggregate	Winter	Without completeness	0.600752	0.873443	84.997	50.003
Aggregate	Winter	Without maturity	0.818716	0.969767	84.997	50.013
Aggregate	Winter	Without agreement	0.742742	0.896971	84.997	50.003
Asset profiles	Summer	Agreement	0.716711	0.949516	85.000	50.018
Asset profiles	Summer	Composite	0.805339	0.915768	85.000	50.008
Asset profiles	Summer	Without freshness	0.793226	0.882145	85.021	50.003
Asset profiles	Summer	Without completeness	0.745745	0.894566	85.021	50.003
Asset profiles	Summer	Without maturity	0.878440	0.984809	85.000	50.003
Asset profiles	Summer	Without agreement	0.785659	0.923676	85.000	50.003
Asset profiles	Winter	Agreement	0.712907	0.947974	84.997	50.003
Asset profiles	Winter	Composite	0.822501	0.920561	84.997	50.003
Asset profiles	Winter	Without freshness	0.801141	0.889122	84.997	50.003
Asset profiles	Winter	Without completeness	0.767625	0.904032	84.997	50.003
Asset profiles	Winter	Without maturity	0.874418	0.984781	84.997	50.008
Asset profiles	Winter	Without agreement	0.839615	0.926409	84.997	50.003

3 Complete controller outcomes

Every rate includes all issued commands, including holds. Costs and losses are mean values per 72-hour block, while violation counts and bus-hours sum all evaluated epochs. Each representation has the same denominator of 46,656 hours per controller.

3.1 IEEE 33

Table 4: Physical, economic and activity summaries for IEEE 33.

Controller	Failures	Rate (%)	Cost (EUR)	Loss (MWh)	Auth. (%)	Full (%)	Active (%)
Ungated	206	0.442	4835.32	0.757	100.00	100.00	39.14
Always half	0	0.000	4815.58	0.509	100.00	0.00	62.52
Metadata	203	0.435	4864.43	0.724	87.52	87.52	34.52

Controller	Failures	Rate (%)	Cost (EUR)	Loss (MWh)	Auth. (%)	Full (%)	Active (%)
Agreement (2)	154	0.330	4849.51	0.721	83.00	83.00	34.36
Composite (2)	202	0.433	4865.47	0.713	83.08	83.08	33.32
Agreement (3)	76	0.163	4851.87	0.630	83.00	48.54	40.70
Composite (3)	85	0.182	4861.54	0.610	83.08	43.54	41.08
Without freshness	85	0.182	4856.52	0.609	82.33	43.28	41.28
Without completeness	80	0.171	4861.08	0.610	83.15	42.74	41.23
Without maturity	85	0.182	4866.56	0.624	83.11	48.80	39.87
Without agreement	90	0.193	4864.88	0.611	83.19	43.20	40.81
Reference forecast	217	0.465	4834.71	0.761	100.00	100.00	38.85
Stress screening	29	0.062	4833.11	0.725	100.00	100.00	40.76
Fixed composite	111	0.238	4848.89	0.657	90.45	60.29	42.02

Table 5: Voltage, thermal and event-extent summaries for IEEE 33.

Controller	$V_{\min}$	$V_{\max}$	$\rho_{\max}$	Mean severity	Longest (h)	Bus-hours
Ungated	0.93671	1.04847	0.2224	0.000011	4	260
Always half	0.96162	1.02811	0.2034	0.000000	0	0
Metadata	0.93671	1.04847	0.2224	0.000011	4	257
Agreement (2)	0.93671	1.04377	0.2224	0.000007	4	192
Composite (2)	0.93671	1.04847	0.2224	0.000010	4	254
Agreement (3)	0.93671	1.04786	0.2224	0.000004	4	99
Composite (3)	0.93671	1.03874	0.2034	0.000005	4	114
Without freshness	0.93671	1.03874	0.2034	0.000005	4	114
Without completeness	0.93671	1.03874	0.2034	0.000005	2	107
Without maturity	0.93671	1.04786	0.2224	0.000005	4	110
Without agreement	0.93671	1.04188	0.2034	0.000006	2	121
Reference forecast	0.93246	1.04941	0.2224	0.000011	4	274
Stress screening	0.94597	1.04847	0.2224	0.000001	1	34
Fixed composite	0.93671	1.04786	0.2224	0.000006	4	144

3.2 MV aggregate

Table 6: Physical, economic and activity summaries for MV aggregate.

Controller	Failures	Rate (%)	Cost (EUR)	Loss (MWh)	Auth. (%)	Full (%)	Active (%)
Ungated	136	0.291	638.35	3.833	100.00	100.00	16.75
Always half	54	0.116	756.06	3.641	100.00	0.00	26.69
Metadata	130	0.279	736.82	3.806	87.52	87.52	14.96
Agreement (2)	88	0.189	671.01	3.805	83.00	83.00	15.07
Composite (2)	116	0.249	727.27	3.804	83.08	83.08	14.95
Agreement (3)	73	0.156	733.21	3.736	83.00	48.54	18.27
Composite (3)	63	0.135	777.18	3.730	83.08	43.54	18.10
Without freshness	62	0.133	759.55	3.729	82.33	43.28	18.09
Without completeness	60	0.129	779.28	3.730	83.15	42.74	18.11
Without maturity	74	0.159	771.29	3.732	83.11	48.80	17.79
Without agreement	62	0.133	793.18	3.732	83.19	43.20	17.59

Controller	Failures	Rate (%)	Cost (EUR)	Loss (MWh)	Auth. (%)	Full (%)	Active (%)
Reference forecast	109	0.234	636.73	3.829	100.00	100.00	16.66
Stress screening	121	0.259	636.57	3.831	100.00	100.00	16.79
Fixed composite	73	0.156	705.06	3.761	90.45	60.29	18.60

Table 7: Voltage, thermal and event-extent summaries for MV aggregate.

Controller	$V_{\min}$	$V_{\max}$	$\rho_{\max}$	Mean severity	Longest (h)	Bus-hours
Ungated	1.00329	1.05425	0.4772	0.000003	2	171
Always half	1.00329	1.05037	0.4126	0.000000	1	54
Metadata	1.00329	1.05425	0.4772	0.000003	2	165
Agreement (2)	1.00329	1.05329	0.4579	0.000001	2	100
Composite (2)	1.00329	1.05394	0.4772	0.000003	2	143
Agreement (3)	1.00329	1.05329	0.4579	0.000001	2	83
Composite (3)	1.00329	1.05329	0.4579	0.000001	2	70
Without freshness	1.00329	1.05329	0.4579	0.000001	2	68
Without completeness	1.00329	1.05329	0.4579	0.000001	2	64
Without maturity	1.00329	1.05329	0.4579	0.000001	2	84
Without agreement	1.00329	1.05329	0.4579	0.000001	2	67
Reference forecast	1.00329	1.05425	0.4892	0.000002	2	134
Stress screening	1.00329	1.05425	0.4772	0.000003	2	147
Fixed composite	1.00329	1.05329	0.4579	0.000001	2	85

3.3 MV asset profiles

Table 8: Physical, economic and activity summaries for MV asset profiles.

Controller	Failures	Rate (%)	Cost (EUR)	Loss (MWh)	Auth. (%)	Full (%)	Active (%)
Ungated	10407	22.306	1072.65	5.493	100.00	100.00	21.65
Always half	9779	20.960	1016.02	5.309	100.00	0.00	36.72
Metadata	10407	22.306	1140.00	5.470	87.57	87.57	19.28
Agreement (2)	10356	22.197	1065.89	5.478	83.30	83.30	19.97
Composite (2)	10382	22.252	1132.58	5.466	82.74	82.74	18.73
Agreement (3)	10146	21.746	1072.14	5.405	83.30	49.69	24.18
Composite (3)	10136	21.725	1097.73	5.405	82.74	44.09	23.47
Without freshness	10063	21.569	1064.97	5.404	80.84	44.11	24.26
Without completeness	10125	21.701	1099.79	5.405	82.75	43.99	23.41
Without maturity	10174	21.806	1114.67	5.405	83.22	49.30	23.32
Without agreement	10099	21.646	1101.79	5.402	82.83	41.40	23.13
Reference forecast	10472	22.445	1064.96	5.494	100.00	100.00	21.00
Stress screening	10235	21.937	1034.93	5.440	100.00	100.00	24.22
Fixed composite	10264	21.999	1068.95	5.460	94.14	77.35	23.30

Table 9: Voltage, thermal and event-extent summaries for MV asset profiles.

Controller	$V_{\min}$	$V_{\max}$	$\rho_{\max}$	Mean severity	Longest (h)	Bus-hours
Ungated	1.00286	1.06222	0.5961	0.001212	55	19555
Always half	1.00469	1.06222	0.5785	0.001159	55	18474
Metadata	1.00286	1.06222	0.5961	0.001213	55	19577
Agreement (2)	1.00286	1.06222	0.5961	0.001207	55	19449
Composite (2)	1.00286	1.06222	0.5961	0.001211	55	19536
Agreement (3)	1.00286	1.06222	0.5961	0.001192	55	19107
Composite (3)	1.00286	1.06222	0.5785	0.001193	55	19120
Without freshness	1.00286	1.06222	0.5785	0.001188	55	18998
Without completeness	1.00286	1.06222	0.5785	0.001195	55	19113
Without maturity	1.00286	1.06222	0.5961	0.001195	55	19172
Without agreement	1.00286	1.06222	0.5785	0.001193	55	19089
Reference forecast	1.00286	1.06438	0.6518	0.001211	55	19632
Stress screening	1.00286	1.06222	0.5785	0.001210	55	19347
Fixed composite	1.00286	1.06222	0.5961	0.001200	55	19330

4 Origin and condition outcomes

Origins and fault conditions are shown separately to expose variation hidden by the overall stress mixture. Seeds are averaged within each physical block. Summer contains eight blocks and winter sixteen; pooled scenario results weight them accordingly.

4.1 IEEE 33

Table 10: Violation rates (%) by origin and condition. A3 and T3 denote agreement and composite three-regime dispatch.

Origin	Condition	Ungated	Half	A3	T3	Reference	Stress
Summer	Clean	0.000	0.000	0.000	0.000	0.000	0.000
Summer	Missing	0.000	0.000	0.000	0.058	0.000	0.000
Summer	Delayed	0.000	0.000	0.000	0.000	0.000	0.000
Summer	Noisy	1.736	0.000	0.058	0.000	0.000	0.058
Summer	Bias	0.000	0.000	0.000	0.000	0.000	0.000
Summer	Disturbance	0.347	0.000	0.289	0.347	0.347	0.000
Summer	Reference outage	0.000	0.000	0.000	0.000	0.000	0.000
Summer	Reference noisy	0.000	0.000	0.116	0.000	1.736	0.000
Summer	Common bias	0.000	0.000	0.000	0.000	0.000	0.000
Winter	Clean	0.260	0.000	0.174	0.260	0.347	0.000
Winter	Missing	0.203	0.000	0.116	0.087	0.347	0.000
Winter	Delayed	0.087	0.000	0.029	0.000	0.347	0.000
Winter	Noisy	1.678	0.000	0.029	0.029	0.347	0.376
Winter	Bias	0.174	0.000	0.087	0.174	0.347	0.000
Winter	Disturbance	1.823	0.000	1.360	1.447	1.389	0.434
Winter	Reference outage	0.260	0.000	0.087	0.174	0.260	0.000
Winter	Reference noisy	0.260	0.000	0.000	0.000	1.678	0.000
Winter	Common bias	0.174	0.000	0.087	0.087	0.174	0.000

4.2 MV aggregate

Table 11: Violation rates (%) by origin and condition. A3 and T3 denote agreement and composite three-regime dispatch.

Origin	Condition	Ungated	Half	A3	T3	Reference	Stress
Summer	Clean	0.347	0.347	0.347	0.347	0.347	0.347
Summer	Missing	0.347	0.347	0.347	0.347	0.347	0.347
Summer	Delayed	0.694	0.347	0.347	0.347	0.347	0.521
Summer	Noisy	1.042	0.347	0.347	0.347	0.347	0.694
Summer	Bias	0.868	0.347	0.347	0.347	0.347	0.694
Summer	Disturbance	0.347	0.347	0.347	0.347	0.347	0.347
Summer	Reference outage	0.347	0.347	0.347	0.347	0.347	0.347
Summer	Reference noisy	0.347	0.347	0.347	0.347	1.042	0.347
Summer	Common bias	0.868	0.347	0.810	0.637	0.868	0.694
Winter	Clean	0.000	0.000	0.000	0.000	0.000	0.000
Winter	Missing	0.000	0.000	0.000	0.000	0.000	0.000
Winter	Delayed	0.000	0.000	0.000	0.000	0.000	0.000
Winter	Noisy	0.463	0.000	0.000	0.000	0.000	0.463
Winter	Bias	0.434	0.000	0.000	0.000	0.000	0.434
Winter	Disturbance	0.000	0.000	0.000	0.000	0.000	0.000
Winter	Reference outage	0.000	0.000	0.000	0.000	0.000	0.000
Winter	Reference noisy	0.000	0.000	0.000	0.000	0.463	0.000
Winter	Common bias	0.434	0.000	0.318	0.116	0.521	0.434

4.3 MV asset profiles

Table 12: Violation rates (%) by origin and condition. A3 and T3 denote agreement and composite three-regime dispatch.

Origin	Condition	Ungated	Half	A3	T3	Reference	Stress
Summer	Clean	16.319	15.104	16.146	15.914	16.493	15.799
Summer	Missing	16.377	15.220	16.204	15.741	16.493	15.799
Summer	Delayed	17.014	15.278	15.509	15.972	16.493	15.799
Summer	Noisy	16.435	15.278	15.220	15.509	16.493	15.799
Summer	Bias	16.840	15.104	16.146	15.914	16.493	15.625
Summer	Disturbance	11.632	10.764	11.516	11.458	11.748	11.285
Summer	Reference outage	16.319	15.104	16.146	16.146	16.551	15.799
Summer	Reference noisy	16.319	15.104	15.336	15.336	16.435	15.799
Summer	Common bias	16.840	15.104	16.551	16.377	16.898	15.625
Winter	Clean	26.215	25.174	26.128	26.186	26.562	26.128
Winter	Missing	26.273	25.174	26.100	25.579	26.562	26.215
Winter	Delayed	26.389	25.260	25.347	26.997	26.562	26.215
Winter	Noisy	27.025	25.347	25.289	25.723	26.562	26.360
Winter	Bias	26.562	24.740	25.926	25.608	26.562	26.389
Winter	Disturbance	17.622	16.146	17.564	17.072	17.911	17.535
Winter	Reference outage	26.215	25.174	25.926	26.042	26.534	26.128
Winter	Reference noisy	26.215	25.174	25.521	25.579	27.025	26.128
Winter	Common bias	26.562	24.740	26.389	25.318	26.678	26.389

5 Paired block uncertainty

Differences are left controller minus right controller. The stratified bootstrap draws eight summer and sixteen winter physical blocks, retaining all conditions and seeds within each selected block, for 10,000 replicates. The percentile intervals are conditional on the sampled benchmark periods. Multiple reported contrasts are descriptive.

5.1 IEEE 33

Table 13: Paired differences and 95% intervals for IEEE 33.

Comparison	Outcome	Estimate	Lower	Upper
T3 minus Ungated	Violation (pp)	-0.2593	-0.3687	-0.1650
T3 minus Ungated	Cost (EUR)	+26.2206	+15.2411	+37.0727
T3 minus Ungated	Loss (MWh)	-0.1475	-0.1578	-0.1370
T3 minus Ungated	Activity (pp)	+1.9397	+1.1703	+2.7285
T3 minus A3	Violation (pp)	+0.0193	-0.0279	+0.0664
T3 minus A3	Cost (EUR)	+9.6746	+4.4834	+14.9238
T3 minus A3	Loss (MWh)	-0.0206	-0.0296	-0.0116
T3 minus A3	Activity (pp)	+0.3879	-0.6623	+1.4339
T2 minus A2	Violation (pp)	+0.1029	+0.0557	+0.1543
T2 minus A2	Cost (EUR)	+15.9577	+9.1012	+23.4712
T2 minus A2	Loss (MWh)	-0.0080	-0.0116	-0.0041
T2 minus A2	Activity (pp)	-1.0395	-1.3804	-0.7073
T3 minus Half	Violation (pp)	+0.1822	+0.1071	+0.2679
T3 minus Half	Cost (EUR)	+45.9662	+29.1661	+63.5060
T3 minus Half	Loss (MWh)	+0.1014	+0.0876	+0.1149
T3 minus Half	Activity (pp)	-21.4378	-22.9252	-19.9567
Half minus Ungated	Violation (pp)	-0.4415	-0.6023	-0.2979
Half minus Ungated	Cost (EUR)	-19.7456	-45.6062	+3.7905
Half minus Ungated	Loss (MWh)	-0.2489	-0.2627	-0.2325
Half minus Ungated	Activity (pp)	+23.3775	+21.6456	+25.1029
Stress minus Half	Violation (pp)	+0.0622	+0.0343	+0.0922
Stress minus Half	Cost (EUR)	+17.5323	-5.3344	+41.5053
Stress minus Half	Loss (MWh)	+0.2163	+0.1988	+0.2323
Stress minus Half	Activity (pp)	-21.7635	-23.2960	-20.1624
Reference minus Ungated	Violation (pp)	+0.0236	-0.0343	+0.0900
Reference minus Ungated	Cost (EUR)	-0.6138	-4.1448	+3.0101
Reference minus Ungated	Loss (MWh)	+0.0037	+0.0002	+0.0069
Reference minus Ungated	Activity (pp)	-0.2915	-0.6687	+0.0922
Stress minus Ungated	Violation (pp)	-0.3794	-0.5316	-0.2486
Stress minus Ungated	Cost (EUR)	-2.2133	-4.3117	-0.3347
Stress minus Ungated	Loss (MWh)	-0.0326	-0.0396	-0.0262
Stress minus Ungated	Activity (pp)	+1.6139	+1.1788	+2.1305

5.2 MV aggregate

Table 14: Paired differences and 95% intervals for MV aggregate.

Comparison	Outcome	Estimate	Lower	Upper
T3 minus Ungated	Violation (pp)	-0.1565	-0.2615	-0.0643
T3 minus Ungated	Cost (EUR)	+138.8311	+92.5076	+185.4938
T3 minus Ungated	Loss (MWh)	-0.1030	-0.1299	-0.0768
T3 minus Ungated	Activity (pp)	+1.3525	+0.7866	+1.9440
T3 minus A3	Violation (pp)	-0.0214	-0.0493	+0.0000
T3 minus A3	Cost (EUR)	+43.9674	+30.9286	+57.5506
T3 minus A3	Loss (MWh)	-0.0052	-0.0101	-0.0003
T3 minus A3	Activity (pp)	-0.1672	-0.6131	+0.2701
T2 minus A2	Violation (pp)	+0.0600	+0.0107	+0.1179
T2 minus A2	Cost (EUR)	+56.2572	+39.1964	+73.7368
T2 minus A2	Loss (MWh)	-0.0012	-0.0074	+0.0048
T2 minus A2	Activity (pp)	-0.1265	-0.4330	+0.1629
T3 minus Half	Violation (pp)	+0.0193	+0.0000	+0.0493
T3 minus Half	Cost (EUR)	+21.1234	+2.9667	+40.6915
T3 minus Half	Loss (MWh)	+0.0889	+0.0659	+0.1125
T3 minus Half	Activity (pp)	-8.5905	-10.7339	-6.4986
Half minus Ungated	Violation (pp)	-0.1758	-0.3044	-0.0664
Half minus Ungated	Cost (EUR)	+117.7077	+70.2958	+166.2922
Half minus Ungated	Loss (MWh)	-0.1919	-0.2393	-0.1432
Half minus Ungated	Activity (pp)	+9.9430	+7.4673	+12.4593
Stress minus Half	Violation (pp)	+0.1436	+0.0514	+0.2551
Stress minus Half	Cost (EUR)	-119.4816	-167.9522	-71.1591
Stress minus Half	Loss (MWh)	+0.1894	+0.1422	+0.2370
Stress minus Half	Activity (pp)	-9.9001	-12.3651	-7.4867
Reference minus Ungated	Violation (pp)	-0.0579	-0.1115	-0.0129
Reference minus Ungated	Cost (EUR)	-1.6175	-13.6123	+10.9990
Reference minus Ungated	Loss (MWh)	-0.0041	-0.0098	+0.0017
Reference minus Ungated	Activity (pp)	-0.0900	-0.3108	+0.1222
Stress minus Ungated	Violation (pp)	-0.0322	-0.0707	+0.0000
Stress minus Ungated	Cost (EUR)	-1.7738	-3.4889	-0.0799
Stress minus Ungated	Loss (MWh)	-0.0025	-0.0039	-0.0014
Stress minus Ungated	Activity (pp)	+0.0429	-0.0257	+0.1372

5.3 MV asset profiles

Table 15: Paired differences and 95% intervals for MV asset profiles.

Comparison	Outcome	Estimate	Lower	Upper
T3 minus Ungated	Violation (pp)	-0.5808	-0.8916	-0.3022
T3 minus Ungated	Cost (EUR)	+25.0790	-16.9155	+65.0431
T3 minus Ungated	Loss (MWh)	-0.0875	-0.1143	-0.0615
T3 minus Ungated	Activity (pp)	+1.8176	+1.0460	+2.6642
T3 minus A3	Violation (pp)	-0.0214	-0.2229	+0.1758
T3 minus A3	Cost (EUR)	+25.5831	-3.3045	+51.0156
T3 minus A3	Loss (MWh)	+0.0004	-0.0086	+0.0081

Comparison	Outcome	Estimate	Lower	Upper
T3 minus A3	Activity (pp)	-0.7116	-1.3996	-0.0214
T2 minus A2	Violation (pp)	+0.0557	-0.0386	+0.1543
T2 minus A2	Cost (EUR)	+66.6888	+45.8449	+89.5811
T2 minus A2	Loss (MWh)	-0.0127	-0.0220	-0.0039
T2 minus A2	Activity (pp)	-1.2367	-1.6182	-0.8723
T3 minus Half	Violation (pp)	+0.7652	+0.4072	+1.1725
T3 minus Half	Cost (EUR)	+81.7122	+42.3034	+118.6886
T3 minus Half	Loss (MWh)	+0.0964	+0.0753	+0.1179
T3 minus Half	Activity (pp)	-13.2523	-16.4138	-10.1316
Half minus Ungated	Violation (pp)	-1.3460	-1.9612	-0.7973
Half minus Ungated	Cost (EUR)	-56.6333	-116.6334	+6.5314
Half minus Ungated	Loss (MWh)	-0.1839	-0.2283	-0.1384
Half minus Ungated	Activity (pp)	+15.0699	+11.2483	+19.0525
Stress minus Half	Violation (pp)	+0.9774	+0.4823	+1.5046
Stress minus Half	Cost (EUR)	+18.9101	-24.3268	+65.6336
Stress minus Half	Loss (MWh)	+0.1311	+0.1008	+0.1621
Stress minus Half	Activity (pp)	-12.4936	-15.6229	-9.5120
Reference minus Ungated	Violation (pp)	+0.1393	-0.0965	+0.4459
Reference minus Ungated	Cost (EUR)	-7.6895	-21.9753	+6.2297
Reference minus Ungated	Loss (MWh)	+0.0008	-0.0027	+0.0045
Reference minus Ungated	Activity (pp)	-0.6494	-1.1167	-0.2379
Stress minus Ungated	Violation (pp)	-0.3687	-0.7416	-0.0193
Stress minus Ungated	Cost (EUR)	-37.7232	-71.0968	-0.4821
Stress minus Ungated	Loss (MWh)	-0.0529	-0.0682	-0.0364
Stress minus Ungated	Activity (pp)	+2.5763	+1.5603	+3.5987

6 Common-proposal selection

Selection uses the same ungated proposals and battery state for every score. Only degraded-condition epochs with a current reference are eligible, giving 40,608 proposals per representation. Ties use seed 1729 and a random ordering independent of realised outcomes. This retrospective selection comparison does not change the deployed development thresholds.

6.1 IEEE 33

Table 16: Common-proposal risk and coverage for IEEE 33.

Score	Coverage (%)	Selected	Failures	Risk (%)
Agreement	50.000	20304	74	0.364
Agreement	75.000	30456	106	0.348
Agreement	90.000	36547	149	0.408
Composite	50.000	20304	90	0.443
Composite	75.000	30456	156	0.512
Composite	90.000	36547	194	0.531
Without freshness	50.000	20304	88	0.433
Without freshness	75.000	30456	127	0.417
Without freshness	90.000	36547	165	0.451
Without completeness	50.000	20304	86	0.424
Without completeness	75.000	30456	155	0.509

Score	Coverage (%)	Selected	Failures	Risk (%)
Without completeness	90.000	36547	194	0.531
Without maturity	50.000	20304	86	0.424
Without maturity	75.000	30456	140	0.460
Without maturity	90.000	36547	191	0.523
Without agreement	50.000	20304	105	0.517
Without agreement	75.000	30456	173	0.568
Without agreement	90.000	36547	194	0.531

6.2 MV aggregate

Table 17: Common-proposal risk and coverage for MV aggregate.

Score	Coverage (%)	Selected	Failures	Risk (%)
Agreement	50.000	20304	42	0.207
Agreement	75.000	30456	62	0.204
Agreement	90.000	36547	76	0.208
Composite	50.000	20304	34	0.167
Composite	75.000	30456	78	0.256
Composite	90.000	36547	117	0.320
Without freshness	50.000	20304	32	0.158
Without freshness	75.000	30456	66	0.217
Without freshness	90.000	36547	94	0.257
Without completeness	50.000	20304	31	0.153
Without completeness	75.000	30456	80	0.263
Without completeness	90.000	36547	117	0.320
Without maturity	50.000	20304	43	0.212
Without maturity	75.000	30456	59	0.194
Without maturity	90.000	36547	117	0.320
Without agreement	50.000	20304	44	0.217
Without agreement	75.000	30456	101	0.332
Without agreement	90.000	36547	118	0.323

6.3 MV asset profiles

Table 18: Common-proposal risk and coverage for MV asset profiles.

Score	Coverage (%)	Selected	Failures	Risk (%)
Agreement	50.000	20304	4308	21.217
Agreement	75.000	30456	6489	21.306
Agreement	90.000	36547	7876	21.550
Composite	50.000	20304	3596	17.711
Composite	75.000	30456	6172	20.265
Composite	90.000	36547	7921	21.673
Without freshness	50.000	20304	3463	17.056
Without freshness	75.000	30456	5933	19.481
Without freshness	90.000	36547	7605	20.809
Without completeness	50.000	20304	3565	17.558

Score	Coverage (%)	Selected	Failures	Risk (%)
Without completeness	75.000	30456	6163	20.236
Without completeness	90.000	36547	7926	21.687
Without maturity	50.000	20304	4266	21.011
Without maturity	75.000	30456	6545	21.490
Without maturity	90.000	36547	8005	21.903
Without agreement	50.000	20304	3383	16.662
Without agreement	75.000	30456	6171	20.262
Without agreement	90.000	36547	7869	21.531

7 Recorded-trajectory stored-energy valuation

The values 90, 130 and 170 EUR/MWh are applied to the recorded terminal-energy difference. Dispatch is not re-optimised. Costs below are mean energy-adjusted values per 72-hour block.

7.1 IEEE 33

Table 19: Mean costs (EUR) under terminal-energy valuations for IEEE 33.

Controller	90 EUR/MWh	130 EUR/MWh	170 EUR/MWh
Ungated	4815.69	4835.32	4854.95
Always half	4789.12	4815.58	4842.04
Metadata	4847.04	4864.43	4881.81
Agreement (2)	4827.78	4849.51	4871.24
Composite (2)	4848.13	4865.47	4882.80
Agreement (3)	4827.52	4851.87	4876.22
Composite (3)	4841.59	4861.54	4881.50
Without freshness	4833.44	4856.52	4879.59
Without completeness	4841.06	4861.08	4881.09
Without maturity	4845.76	4866.56	4887.36
Without agreement	4844.76	4864.88	4885.00
Reference forecast	4814.68	4834.71	4854.74
Stress screening	4811.02	4833.11	4855.20
Fixed composite	4828.42	4848.89	4869.35

7.2 MV aggregate

Table 20: Mean costs (EUR) under terminal-energy valuations for MV aggregate.

Controller	90 EUR/MWh	130 EUR/MWh	170 EUR/MWh
Ungated	676.93	638.35	599.76
Always half	792.81	756.06	719.30
Metadata	771.16	736.82	702.49
Agreement (2)	706.65	671.01	635.38
Composite (2)	766.05	727.27	688.49

Controller	90 EUR/MWh	130 EUR/MWh	170 EUR/MWh
Agreement (3)	772.65	733.21	693.77
Composite (3)	817.55	777.18	736.81
Without freshness	802.37	759.55	716.74
Without completeness	818.90	779.28	739.66
Without maturity	804.92	771.29	737.66
Without agreement	827.66	793.18	758.70
Reference forecast	675.85	636.73	597.61
Stress screening	675.29	636.57	597.86
Fixed composite	745.61	705.06	664.51

7.3 MV asset profiles

Table 21: Mean costs (EUR) under terminal-energy valuations for MV asset profiles.

Controller	90 EUR/MWh	130 EUR/MWh	170 EUR/MWh
Ungated	1099.74	1072.65	1045.56
Always half	1033.70	1016.02	998.34
Metadata	1164.40	1140.00	1115.60
Agreement (2)	1089.37	1065.89	1042.42
Composite (2)	1156.80	1132.58	1108.37
Agreement (3)	1095.56	1072.14	1048.73
Composite (3)	1119.81	1097.73	1075.64
Without freshness	1091.36	1064.97	1038.58
Without completeness	1121.10	1099.79	1078.47
Without maturity	1133.91	1114.67	1095.43
Without agreement	1123.25	1101.79	1080.34
Reference forecast	1091.37	1064.96	1038.55
Stress screening	1068.76	1034.93	1001.09
Fixed composite	1094.91	1068.95	1042.99

8 Retrospective action feasibility

Thirteen tested power fractions are -1, -0.75, -0.5, -0.375, -0.25, -0.125, 0, 0.125, 0.25, 0.375, 0.5, 0.75 and 1. AC feasibility is evaluated at realised inputs; energy eligibility uses the controller's recorded starting state of charge. Each of 6,912 unique realised states admits at least one tested power before energy screening. Energy-limited and alternative-available counts partition all observed failures because the no-feasible-power category is zero. No feasible issued command is missed by the diagnostic. Alternative availability is a same-hour comparison with a fixed prior trajectory.

8.1 IEEE 33

Table 22: Failure partition and forecast-infeasible holds for IEEE 33.

Controller	Failures	Energy-limited	Alternative	Model hold	Hold failure
Ungated	206	0	206	0	0
Always half	0	0	0	0	0
Metadata	203	0	203	0	0
Agreement (2)	154	0	154	0	0
Composite (2)	202	0	202	0	0
Agreement (3)	76	0	76	0	0
Composite (3)	85	0	85	0	0
Without freshness	85	0	85	0	0
Without completeness	80	0	80	0	0
Without maturity	85	0	85	0	0
Without agreement	90	0	90	0	0
Reference forecast	217	0	217	0	0
Stress screening	29	0	29	0	0
Fixed composite	111	0	111	0	0

8.2 MV aggregate

Table 23: Failure partition and forecast-infeasible holds for MV aggregate.

Controller	Failures	Energy-limited	Alternative	Model hold	Hold failure
Ungated	136	0	136	0	0
Always half	54	0	54	0	0
Metadata	130	0	130	0	0
Agreement (2)	88	0	88	0	0
Composite (2)	116	0	116	0	0
Agreement (3)	73	14	59	0	0
Composite (3)	63	0	63	0	0
Without freshness	62	4	58	0	0
Without completeness	60	2	58	0	0
Without maturity	74	10	64	0	0
Without agreement	62	0	62	0	0
Reference forecast	109	0	109	0	0
Stress screening	121	0	121	0	0
Fixed composite	73	0	73	0	0

8.3 MV asset profiles

Table 24: Failure partition and forecast-infeasible holds for MV asset profiles.

Controller	Failures	Energy-limited	Alternative	Model hold	Hold failure
Ungated	10407	8186	2221	8313	7423
Always half	9779	6943	2836	7901	7118
Metadata	10407	7245	3162	8348	7455
Agreement (2)	10356	8031	2325	8301	7423
Composite (2)	10382	7132	3250	8343	7452
Agreement (3)	10146	7763	2383	8193	7355
Composite (3)	10136	6610	3526	8209	7364
Without freshness	10063	7287	2776	8150	7305
Without completeness	10125	6555	3570	8215	7367
Without maturity	10174	6955	3219	8244	7387
Without agreement	10099	6517	3582	8196	7353
Reference forecast	10472	8244	2228	8490	7758
Stress screening	10235	7694	2541	8482	7565
Fixed composite	10264	7898	2366	8246	7375

9 Numerical audit and archive structure

Arithmetic checks cover every record. Fresh uncached AC checks cover sampled and failed operating points in each origin and representation. Profile mapping is independently checked against official SimBench absolute-value conversion over all 8,784 benchmark hours.

Table 25: Saved-record and fresh-solve audit.

Check	Value
Decision count	1959552
Maximum soc error	1.1102e-16
Soc bounds errors	0
Nonzero model infeasible	0
Nonfinite outcomes	0
Proposal battery bounds errors	0
Maximum command rule error	5.5511e-17
Maximum score error	3.3307e-16
Physical label disagreements	0
Regime disagreements	0
Reference fallback disagreements	0
Fresh ac samples	114
Fresh model ac samples	114
Maximum fresh ac difference	6.2493e-08
Fresh ac label disagreements	0
Fresh model label disagreements	0

File checksums and the study and diagnostic source hashes accompany the archive in its manifests and metadata.